[Project Title]

[A subtitle naming the ML task and visualization goal]

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Blueprint: the topic is undecided. Replace every bracketed prompt and email placeholder with verified project details before submission.

1 Problem, Motivation, and Research Question

Problem and intended users. *[Identify a specific user group, the data they work with, and a decision or analysis task they currently find difficult. Describe the present workflow and its concrete limitation.]*

Why this matters. *[Explain the consequence of that limitation: missed errors, weak understanding, slow analysis, or another observable cost. Give a realistic usage scenario and cite evidence where available.]*

Primary research question. Can *[specific visualization or interaction]* help *[target users]* *[perform a measurable task involving an ML model or its outputs]* more effectively than *[a named baseline]*, on *[dataset/domain]*?

Working hypothesis. We hypothesize that *[design choice]* will improve *[primary outcome]* because *[a plausible mechanism]*. We will test this against *[comparison condition]*; the result may support or reject the hypothesis.

Project in one paragraph. We propose to build *[a specific artifact: interactive tool, visual analytics system, or visualization-assisted ML workflow]*. It will take *[inputs]*, use *[ML method]*, and present *[linked visual views and interactions]*. Users will be able to *[two or three concrete actions]*. Its intended contribution is *[one focused improvement]*, evaluated through *[comparison and metric]*.

Scope boundary. *[Define one primary task and a minimum working demonstration. State a practical limit on model size, datasets, or supported interactions so the project is feasible.]*

2 Key Research Paper and Critical Analysis

Key paper. *[Authors. Exact title. Venue, year. DOI or stable URL. Add a verified BibTeX entry to references.bib and cite it here.]*

Main Ideas, Methods, and Contributions

Research objective and central idea. *[In your own words, explain the problem the paper addresses, its intended users, and its central insight. Explain why this paper is foundational to your proposed topic.]*

Method and system. *[Describe the data, ML technique, visual encodings, and interactions. Explain the information flow from model or data to visualization and user action. Include enough detail to show how the method works.]*

Contributions and evidence. *[Identify the authors' main contributions. Summarize their evaluation design, datasets or participants, baselines, and relevant outcomes. Cite the relevant section, figure, or table for any specific result.]*

Critical Assessment

Dimension	Analysis to complete
Strength	*[Which design or method works well, and what evidence supports that judgment?]*
Limitation	*[Which assumption, usability issue, scale constraint, or evaluation weakness limits the work? Distinguish reported limitations from your interpretation.]*
Unanswered question	*[What relevant question remains unresolved, and why does the paper's evidence not answer it?]*
Project implication	*[Which limitation can your team realistically investigate, and what result would address it?]*

What We Will Build Upon or Extend

Retain or adapt. *[Name the exact concept, algorithm, visual encoding, or interaction you will reuse. Explain how it fits the proposed task and whether code or model artifacts are available.]*

Extend and test. *[State a specific change to the paper's approach. Connect that change to the limitation above and identify the comparison that will isolate its benefit.]*

Related work beyond the key paper. *[Briefly position two or three additional sources: a relevant visualization approach, an ML method or baseline, and an evaluation method as appropriate. Explain what each contributes to the project and add citations.]*

3 Proposed Project and Innovation

System and User Workflow

What we intend to build. *[Describe the working artifact, its intended user, its inputs and outputs, and the single task it must support end to end. Specify whether the ML model is pretrained, trained by the team, or updated through user interaction.]*

1. **Load and prepare:** *[What data enters the system? What preprocessing and model computation occur?]*
2. **Inspect:** *[What does the user see first, and which visual encoding reveals the relevant patterns?]*
3. **Interact:** *[Specify selection, filtering, linked views, comparison, or another interaction and what updates as a result.]*
4. **Act and verify:** *[What decision or insight can the user produce, and how can they inspect supporting evidence?]*

Connection Between Visualization and Machine Learning

Component	Design commitment
ML task and method	*[Task, candidate model, inputs, and outputs; justify the simplest suitable method.]*
Visual representation	*[Map model/data attributes to marks, position, color, or other encodings; explain the choice.]*
User interaction	*[Specify how user actions change a view, a comparison, or the model. Clarify whether retraining occurs.]*
Analytical benefit	*[Explain what a user can learn or do that raw predictions or a static table do not readily support.]*

Innovation and Comparison With Existing Work

Closest existing approach. *[Name and cite the closest system or method. Describe how it would handle the same task and where the proposed workflow differs.]*

Expected contribution. Our proposed contribution is *[a precise design, method, adaptation, or empirical finding]*, distinguished by *[one or two concrete differences]*. We will assess its value by *[comparison that tests those differences]*. *[Explain why the contribution is useful even if the underlying ML algorithm is established. Avoid claiming novelty without checking the literature.]*

Minimum deliverable and optional extension. *[Define a working core with one dataset, one ML method, and the essential views/interactions. Identify one optional extension to attempt only after the core and evaluation work.]*

4 Project Plan and Evaluation

Dataset and Access

[Name the primary dataset, provider, stable access link, license, approximate size, features, and labels if needed. Explain how it will be downloaded or requested, preprocessing, and the unit of analysis. Name a readily available fallback.]

Implementation and Feasibility

[Specify the model/library, visualization framework, data interface, and compute needs. Explain separation of training, validation, and test data if training is involved, reproducibility (versions and seeds), and how sensitive fields or misleading visual summaries will be handled if relevant. Cite methods and data sources.]

Milestones and Provisional Ownership

Confirm roles together; add dates after the course timeline is known.

Stage	Lead	Completion evidence / target date
Scope	All	Selected paper, question, dataset access; by Oct. 20.
Data + ML		*[Runnable preprocessing and baseline; date.]*
Visual design		*[Task-driven sketches and interactive core; date.]*
Integration		*[End-to-end model/view pipeline; date.]*
Evaluation		*[Pilot, comparison results, and analysis; date.]*
Delivery	All	*[Reproducible demo, report, and limitations; date.]*

Evaluation Design

[Define representative tasks, a baseline condition, and the same data and task difficulty across conditions. Choose a feasible participant study, expert review, or reproducible benchmark and justify it. If using participants, specify recruitment, target count, ordering/counterbalancing, and a pilot.]

Measures and Success Criteria

[Choose a primary outcome tied to the research question (e.g., task correctness or error discovery), a secondary measure (e.g., completion time or usability), and a justified target. For ML performance, select a task-appropriate metric and held-out protocol. Specify how comparisons, uncertainty, and qualitative feedback will be analyzed; a small exploratory study should not imply broad generalization.]

Risks and Fallback

[Name the most likely data, compute, integration, or recruitment failure and its fallback. State what can still be evaluated if the optional extension or user study is infeasible, and what conclusions that fallback cannot support.]

References

[Add and cite the key paper, related work, and dataset source using references.bib. Verify all entries and resolve all citations before submission.]